A

Project Report

On

CHAT UI WEBSITE

Submitted during 5thsemesterinpartialfulfilmentoftherequirements for the award of degree of

Bachelor of Technology

in

Electronics & Computer Engineering

by

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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in this Project titled " _ Chat UI " in fulfilment of the requirements for the degree of Bachelor of Technology in Computer engineering and submitted to "J.C. Bose University of Science and Technology, YMCA, Faridabad", is an authentic record of my own work carried out under the supervision of Ms. Meenakshi .

Student's Signature

Student's Name

Ansh Singh

CERTIFICATE

It is hereby certified that the project titled Chat UI Website submitted by Ansh Singh (23001015008) of Btech,ENC-3rd year(5th semester) to "**J. C. Bose University of Science and Technology, YMCA, Faridabad**" for the award of the degree of Bachelor of Technology in Computer Engineering is a record of a bonafide work carried out by him/her in the Project Lab – J.C. Bose University of Science and Technology, YMCA, Faridabad under my supervision as mentor for December 2025 examination.

Ms. Meenakshi

Assistant Professor

Department of Computer Engg.

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I would like to extend my sincere thanks to all them. I am highly indebted to my mentor Ms. Meenakshi for their guidance and constant supervision as well as for providing necessary information regarding the project & also for their support in completing the project.

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Student's Signature

Student's Name
Ansh Singh

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CHAPTER 1: INTRODUCTION

1.1 BRIEF OF THE PROJECT

Brief of the Project The "Real-Time Web Chat Application" is a browser-based communication platform designed to facilitate instant messaging and video conferencing. Built to be lightweight and responsive, the application allows users to register via email, exchange text messages in real-time with zero latency, share multimedia files (images/documents), and initiate secure 1-on-1 video calls. The system ensures user privacy through authentication and offers a modern user interface similar to popular social media platforms.

1.2 TECHNOLOGIES USED

Frontend Interface: HTML5, CSS3 (using Tailwind CSS framework for styling).

Logic & Interactivity: JavaScript (ES6 Modules).

Backend Service: Google Firebase (Backend-as-a-Service).

Authentication: Firebase Auth (Email/Password login & Signup).

Database: Cloud Firestore (NoSQL real-time database for storing chats and user data).

File Storage: Firebase Storage (For uploading profile pictures and shared files).

Video Calling: Jitsi Meet API (External integration for WebRTC video calls).

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The evolution of web-based communication has shifted drastically from static, asynchronous models to dynamic, real-time interactions.

Early web chat systems relied on HTTP Polling, where the client would repeatedly request data from the server at fixed intervals. This method was inefficient, causing high server load and latency.

Modern applications have adopted WebSockets and Server-Sent Events (SSE), which allow for a persistent connection between the client and server. This project leverages Google Firebase, which abstracts these complex socket connections to provide instantaneous data synchronization across clients.

CHAPTER 3: IMPLEMENTATION AND RESULT ANALYSIS

3.1 PROBLEM FORMULATION

In an increasingly digital world, the need for instant communication tools is paramount. However, building a scalable chat backend from scratch is complex and resource-intensive. Furthermore, integrating video calling often requires separate software. This project addresses these issues by utilizing Serverless architecture (Firebase) and API integrations (Jitsi) to create a unified communication tool.

3.2 OBJECTIVES

To implement a secure User Authentication system (Sign Up/Login/Logout).

To achieve Real-Time Synchronization of messages between users without page reloads.

To integrate Multimedia Support allowing users to upload and view images/files.

To implement User Management features like blocking contacts and profile customization.

To seamlessly integrate Video Conferencing within the chat interface.

3.3 METHODOLOGY

The project was developed using an Agile methodology:

UI Design: The layout was crafted using Tailwind CSS to ensure mobile responsiveness.

Firebase Configuration: A Firebase project was set up to handle Auth, Firestore, and Storage buckets.

Core Logic Development: JavaScript modules were written to handle event listeners (sending messages, file inputs).

Integration: The Jitsi Meet External API was embedded to handle video streams via an overlay modal.

Testing: The application was tested for latency, file upload limits, and cross-device compatibility.

CHAPTER 4: APPLICATIONS AND SCOPE

1.1 APPLICATIONS

1. Student Collaboration: Can be used by students (like in the ECE department) to share project files and discuss code.
2. Remote Assistance: Useful for troubleshooting where video and file sharing are needed simultaneously.
3. Internal Organization Tool: Can be deployed as a private communication tool for small organizations or clubs.

1.2 FUTURE SCOPE

1. Group Chat: Extending the current 1-on-1 logic to support multi-user rooms.
2. End-to-End Encryption: Implementing client-side encryption for enhanced privacy.
3. Mobile App Conversion: Using frameworks like Ionic or React Native to convert this web app into a standalone Android/iOS app.
4. AI Chatbot: Integrating an AI API to provide automated answers within the chat..

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